10/3/2025

lunedì 10 marzo 2025 11:18

$\bar{x} \tilde{=} x$

Errore assoluto $ERR_A \overset{def}{=} |x-\bar{x}|$

ERRORE RELATIVO $ERR_R \overset{def}{=} \dfrac{|x-\bar{x}|}{|x|}$

$x=20 \quad \bar{x}=10 \quad E_A = |x-\bar{x}| = 10 \quad E_R = \frac{10}{20} = 0{,}5$

$x = 1000 \quad \bar{x} = 990 \quad E_A = |x-\bar{x}| = 10 \quad E_R = \frac{10}{1000} = 0{,}01$

$a = p \cdot N^q \qquad \frac{1}{N} \leq |p| < 1 \qquad \bar{a} = \bar{p} \cdot N^q$

t = numero di cifre a disposizione per $|p|$

ARROTONDAMENTO
$$\boxed{|p-\bar{p}| \leq \frac{1}{2} N^{-t}}$$

moltiplicando 1° e 2° membro per N^q

TEOREMA

$$\frac{|a-\bar{a}|}{|a|} \leq \frac{1}{2} \cdot N^{1-t} \quad) \text{ eps}$$

DIM

$$|a-\bar{a}| = |p \cdot N^q - \bar{p} \cdot N^q| = |p-\bar{p}| \cdot N^q \leq \frac{1}{2} N^{-t} \cdot N^q$$

$$\Rightarrow |a-\bar{a}| \leq \frac{1}{2} \cdot N^{q-t} \quad (1)$$

$$|a| = |p \cdot N^q| = |p| \cdot N^q \geq \frac{1}{N} \cdot N^q \Rightarrow |a| \geq N^{q-1}$$

$$\Rightarrow \frac{1}{|a|} \leq \frac{1}{N^{q-1}} \quad (2)$$

MOLTIPLICANDO MEMBRO A MEMBRO (1) * (2):

$$\frac{|a-\bar{a}|}{|a|} \leq \frac{1}{2} \cdot N^{q-t} \cdot \frac{1}{N^{q-1}} \Rightarrow \frac{|a-\bar{a}|}{|a|} \leq \frac{1}{2} N^{\cancel{q}-t-\cancel{q}+1}$$

C.V.D. ■

ESEMPIO SOMMA

$b_1 = 0.2841*10^{-3}, \qquad b_2 = 0.1248*10^{-7}$

$b_2 = 0.00001248*10^{-3} \qquad N=10 \qquad t=4$

$b_1 + b_2 = 0.28410248 \cdot 10^{-3}$

$b_1 \oplus b_2 = 0.2841 \cdot 10^{-3} \neq b_1$

CARTA e penna

$b_3 = 0.1248 \cdot 10^{-6}$

$b_3 = 0.0001248 \cdot 10^{-3}$

$b_1 + b_3 = 0.2842248 \cdot 10^{-3}$

$b_1 \oplus b_3 = 0.2842 \cdot 10^{-3}$

$\neq b_1$

$b_1 \oplus b_2 = 0.2842 \cdot 10 \neq b_1$

$b_1 \oplus b_3 = 0.2842 \cdot 10^{-3} \neq b_1$

$q_2 - q_1 = 3 < t$

CARTA e penna

$a + b = a \iff b = 0$

Su calcolatore può accadere che:

$a \oplus b = a$ con $b \neq 0$

Quando?

$a = p_1 \cdot N^{q_1}$ $b = p_2 \cdot N^{q_2}$

$q_2 < q_1$

nell'esempio precedente $q_1 = -3$ $q_2 = -7$

$-3-(-7) = 4 \neq t$

$a \oplus b = a$ quando $q_1 - q_2 > t$

$a \oplus b \neq a$ quando $q_1 - q_2 < t$

quando $q_1 - q_2 = t$ dipende da "P_{t+1} di b"

$t = 16$ $N = 10$

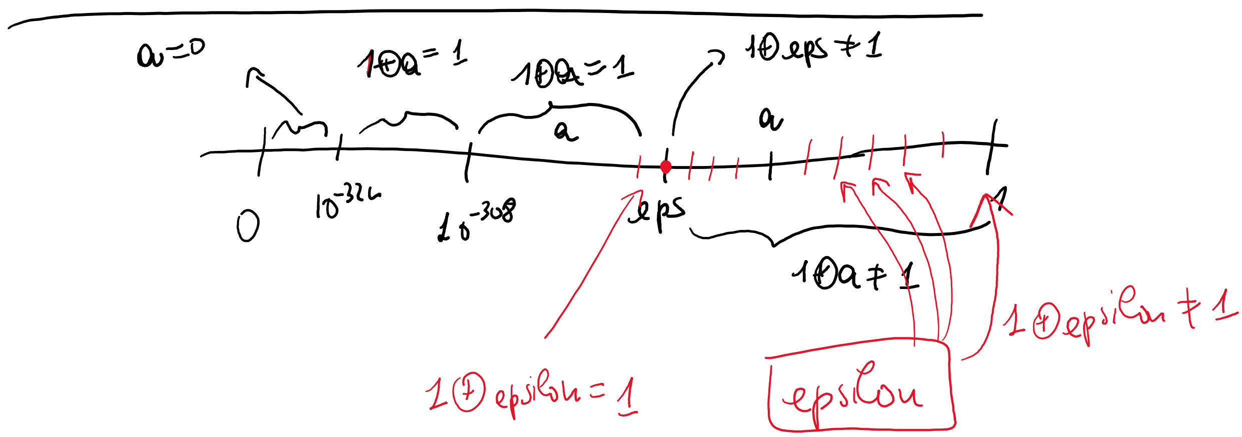
```
Command Window
New to MATLAB? See resources for Getting Started.
>> a=100;
>> b=1e50;
>> c=a+b

c =

   1.0000e+50

>> c==b

ans =

  logical

   1

fx >>
```

ESERCIZI DALLE SLIDES

```
>> a=1;
>> b=0.001;
>> a+b

ans =

   1.0010

>> a=1;
>> b=0.000001;
>> a+b

ans =

   1.0000

>> c=a+b

c =

   1.0000

>> format long
>> c==a

ans =
```

```
  logical

   0

>> c

c =

   1.000001000000000

>> a=1; b=1e-12;
>> c=a+b

c =

   1.000000000001000

>> a=1; b=eps;
>> c=a+b

c =

   1.000000000000000

>> c==a

ans =

  logical

   0

>> a=1; b=1e-17;
>> a+b

ans =

     1

>> ans==a

ans =

  logical

   1

>> a=1e6; b=1e-12;
>> c=a+b

c =

     1000000

>> c==a

ans =

  logical

   1

>> a=1e6; b=1e-11;
>> a+b

ans =

     1000000

>> eps

ans =

     2.220446049250313e-16

>>
```

$a = 0$ $\quad 1 \oplus a = 1 \quad 1 \oplus a = 1 \quad 1 \oplus eps \neq 1$



epsilon = 1
wile 1⊕epsilon ≠ 1
epsilon = epsilon/2
end
epsilon = epsilon * 2